CPE 486/586: Deep Learning for Engineering Applications

12 Neural Ordinary Differential Equations

Spring 2026

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Outline

Ordinary Differential Equations

Initial Value Problems

$$\frac{dx(t)}{dt} = f(x(t), t, \theta); \quad x(t_0) \text{ is given}; \quad x(t_1) = ?$$

- x is a variable we are interested in,
- t is (typically) time,
- f is a function of x and t; it is the differential,
- θ parameterizes f (optionally).

Many physical processes follow this template!

Solution:

$$x(t_1) = x(t_0) + \int_{t_0}^{t_1} f(x(t), t, \theta)\, dt$$

Example

$$\frac{dx}{dt} = 2t; \quad x(0) = 2; \quad x(1) = ?$$

$$\begin{aligned}
\Rightarrow x(1) &= x(0) + \int_0^1 2t\, dt \\
&= x(0) + \left(t^2\big|_{t=1} - t^2\big|_{t=0}\right) \\
&= 2 + 1^2 - 0^2 = 3
\end{aligned}$$

Ordinary Differential Equations (ODEs)

$$x(t_1) = x(t_0) + \boxed{\int_{t_0}^{t_1} f(x(t), t, \theta)\, dt} \quad \longrightarrow$$ What if this cannot be analytically integrated?

Example

$$\frac{dx}{dt} = 2xt; \quad x(0) = 3$$

$$\Rightarrow \int \frac{1}{2x}\, dx = \int t\, dt$$

$$\Rightarrow \tfrac{1}{2} \log x = \tfrac{1}{2} t^2 + c_0$$

$$\Rightarrow x(t) = ce^{t^2}, \quad x(0) = 3 \Rightarrow c = 2$$

$$\therefore x(t) = 2e^{t^2} \Rightarrow x(1) = 5.436$$

Ordinary Differential Equations (ODEs)

$$x(t_1) = x(t_0) + \int_{t_0}^{t_1} f(x(t), t, \theta)\, dt$$

Approximations to $\int_{t_0}^{t_1} f(x(t), t, \theta)\, dt$ i.e. Numerical Integration

- Euler method
- Runge-Kutta methods
- ...

Ordinary Differential Equations (ODEs)

Initial value problem:

$$\frac{dx(t)}{dt} = f(x(t), t, \theta); \quad x(t_0) \text{ is given}; \quad x(t_1) = ?$$

Solution:

$$x(t_1) = x(t_0) + \int_{t_0}^{t_1} f(x(t), t, \theta)\, dt$$

1st-order Runge-Kutta / Euler's method

$$t_{n+1} = t_n + h \quad \longrightarrow \text{ Step size } h$$

$$x(t_{n+1}) = x(t_n) + h f(x(t_n), t_n) \quad \longrightarrow \text{ Update using derivative } f$$

Step size matters!

Ordinary Differential Equations (ODEs) – Euler Example

1st-order Runge-Kutta / Euler's method:

$$t_{n+1} = t_n + h$$
$$x(t_{n+1}) = x(t_n) + hf(x(t_n), t_n)$$

Example: $\frac{dx}{dt} = f(x,t) = 2xt$; $x(0) = 3$; $x(1) = ?$ **(True:** $x(1) \approx 5.436$**)**

$$h = 0.25$$
$$x(0.25) = x(0) + 0.25 \cdot f(x(0), 0) = 3 + 0.25 \cdot (2 \cdot 3 \cdot 0) = 3$$
$$x(0.5) = x(0.25) + 0.25 \cdot f(x(0.25), 0.25) = 3 + 0.25 \cdot (2 \cdot 3 \cdot 0.25) = 3.375$$
$$x(0.75) = x(0.5) + 0.25 \cdot f(x(0.5), 0.5) = 3.375 + 0.25 \cdot (2 \cdot 3.375 \cdot 0.5) = 4.21875$$
$$x(1) = x(0.75) + 0.25 \cdot f(x(0.75), 0.75)$$
$$= 4.21875 + 0.25 \cdot (2 \cdot 4.21875 \cdot 0.75) \approx 5.8008$$

Ordinary Differential Equations (ODEs)

★ 4th-order Runge-Kutta method:

$$
\begin{aligned}
t_{n+1} &= t_n + h \\
s_1 &= f(x(t_n),\ t_n) \\
s_2 &= f\left(x(t_n) + \tfrac{h}{2} s_1,\ t_n + \tfrac{h}{2}\right) \\
s_3 &= f\left(x(t_n) + \tfrac{h}{2} s_2,\ t_n + \tfrac{h}{2}\right) \\
s_4 &= f(x(t_n) + h\, s_3,\ t_n + h) \\
x(t_{n+1}) &= x(t_n) + \frac{h}{6}(s_1 + 2s_2 + 2s_3 + s_4)
\end{aligned}
$$

--→ Default ODE solver used in MATLAB.

Ordinary Differential Equations (ODEs)

Solution as an ODE solver call:

$$x(t_1) = \text{ODESolve}\Big(\underbrace{f(x(t), t, \theta)}_{\text{Differential}}, \underbrace{x(t_0)}_{\text{Initial value}}, \underbrace{t_0}_{\text{Initial time}}, \underbrace{t_1}_{\text{Final time}}\Big)$$

Any ODE solver of our choice!

Ordinary Differential Equations (ODEs)

Fundamental Theorem of ODEs

Suppose f is continuously differentiable. Then:

1. The solution curves for the differential equation completely fill the plane (geometrically, $x(t)$ is a *flow*!), and
2. Solution curves for different solutions do not intersect.

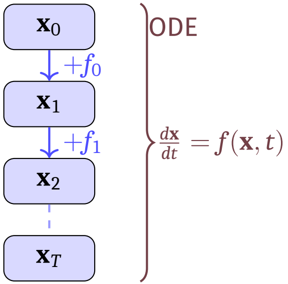
ODEs in the Context of Discrete Dynamical Systems

Deep Networks as Discrete Dynamical Systems

- Modern deep networks (ResNets, DenseNets) stack many layers
- Each layer transforms a hidden state:

$$\mathbf{x}_{t+1} = \mathbf{x}_t + f(\mathbf{x}_t,\, \theta_t)$$

- This is **Euler's method** applied to an ODE!
- As layer count $\to \infty$, the discrete steps become a continuous flow



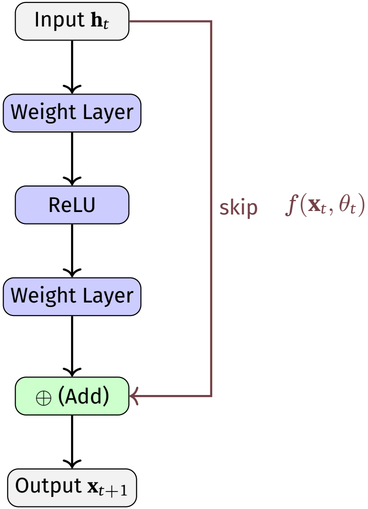
What if we *directly* model the continuous dynamics instead of discretizing them by hand?

Residual Networks (ResNets)

ResNet Building Block

$$\mathbf{x}_{t+1} = \mathbf{x}_t + f(\mathbf{x}_t, \theta_t), \quad t = 0, 1, \ldots, T-1$$

- $\mathbf{x}_t \in \mathbb{R}^d$: hidden state at layer t
- f: neural network (residual function)
- Skip connection ensures gradient flow
- Enables very deep networks (100+ layers)



Connection to Euler Method

Euler's method for $\frac{d\mathbf{x}}{dt} = f(\mathbf{x}, t)$ with step $\Delta t = 1$:

$$\mathbf{x}(t + \Delta t) \approx \mathbf{x}(t) + \Delta t \cdot f(\mathbf{x}(t), t)$$

From Discrete to Continuous Depth

As number of layers $T \to \infty$ and step size $\Delta t \to 0$, the residual update

$$\mathbf{x}_{t+1} = \mathbf{x}_t + \Delta t f(\mathbf{x}_t,\, t,\, \theta)$$

converges to the **continuous ODE**:

$$\boxed{\frac{d\mathbf{x}(t)}{dt} = f(\mathbf{x}(t),\, t,\, \theta)}$$

Discrete: ResNet

$$\mathbf{x}_T = \mathbf{x}_0 + \sum_{t=0}^{T-1} f(\mathbf{x}_t, \theta_t)$$

Continuous: Neural ODE

$$\mathbf{x}(T) = \mathbf{x}(0) + \int_0^T f(\mathbf{x}(t), t, \theta)\, dt$$

The parameters θ are **shared across all depths**, unlike ResNets with per-layer weights. This gives **constant parameter count** regardless of integration depth.

Ordinary Differential Equations (ODEs)

Initial Value Problem (IVP)

Given dynamics f and initial condition $\mathbf{x}(t_0) = \mathbf{x}_0$, find $\mathbf{x}(t)$:

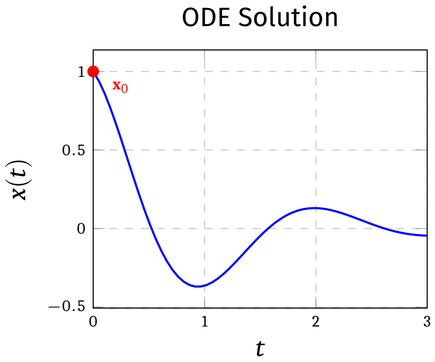
$$\frac{d\mathbf{x}}{dt} = f(\mathbf{x}(t), t), \quad \mathbf{x}(t_0) = \mathbf{x}_0$$

Solution (via Picard–Lindelöf):

$$\mathbf{x}(t) = \mathbf{x}_0 + \int_{t_0}^{t} f(\mathbf{x}(s), s)\, ds$$

Existence & Uniqueness

If f is **Lipschitz continuous** in $\mathbf{h}$, a unique solution exists on $[t_0, T]$.



Numerical ODE Solvers

Euler Method (1st Order)

$$\mathbf{x}_{t+h} \approx \mathbf{x}_t + hf(\mathbf{x}_t, t)$$

Simple but low accuracy; $\mathcal{O}(h)$ error per step.

Runge-Kutta 4 (RK4)

$$\begin{aligned}
k_1 &= f(\mathbf{x}_t,\, t) \\
k_2 &= f\left(\mathbf{x}_t + \tfrac{h}{2}k_1,\, t + \tfrac{h}{2}\right) \\
k_3 &= f\left(\mathbf{x}_t + \tfrac{h}{2}k_2,\, t + \tfrac{h}{2}\right) \\
k_4 &= f(\mathbf{x}_t + h\,k_3,\, t + h) \\
\mathbf{x}_{t+h} &= \mathbf{x}_t + \tfrac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4)
\end{aligned}$$

$\mathcal{O}(h^4)$ error per step.

Adaptive Solvers (Dopri5)

- Embed two RK schemes of different orders
- Estimate local truncation error
- **Increase** step size when error is small
- **Decrease** step size when error is large
- Control solution to a user-specified tolerance: `rtol`, `atol`

Neural ODE dynamics can vary rapidly in some regions; adaptive solvers allocate computation where needed, yielding efficiency and accuracy.

Introducing Neural ODEs

Definition

A **Neural ODE** parametrizes the derivative of a hidden state with a neural network f_θ:

$$\frac{d\mathbf{x}(t)}{dt} = f_\theta(\mathbf{x}(t),\, t)\,, \qquad \mathbf{x}(t_0) = \mathbf{x}_0$$

The output at time t_1 is computed by **numerically integrating** the ODE:

$$\mathbf{x}(t_1) = \mathbf{x}(t_0) + \int_{t_0}^{t_1} f_\theta(\mathbf{x}(t), t)\, dt \;=\; \texttt{ODESolve}(f_\theta,\, \mathbf{x}_0,\, t_0,\, t_1)$$

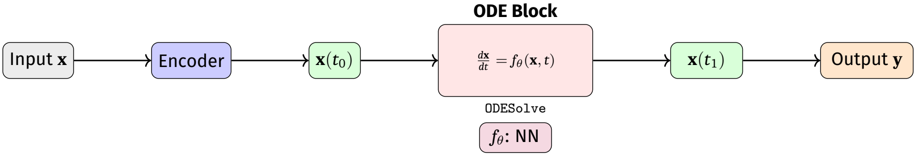
Forward Pass

1. Set initial state $\mathbf{x}(t_0)$ (e.g., encoder output)
2. Call ODE solver with f_θ
3. Obtain $\mathbf{x}(t_1)$ as the model output
4. Apply output layer / loss

Backward Pass

1. Do **NOT** backprop through solver steps
2. Solve a **reverse-time ODE** (adjoint)
3. Compute gradients $\nabla_\theta \mathcal{L}$ efficiently
4. Memory cost: $\mathcal{O}(1)$ in depth

Neural ODE: Architecture View



Continuous Depth

The "depth" of the network is the integration interval $[t_0, t_1]$. More complex inputs naturally require more solver steps (adaptive depth).

Parameter Efficiency

f_θ is **shared** across all integration steps. Total parameters determined by the architecture of f_θ, *not* by integration depth.

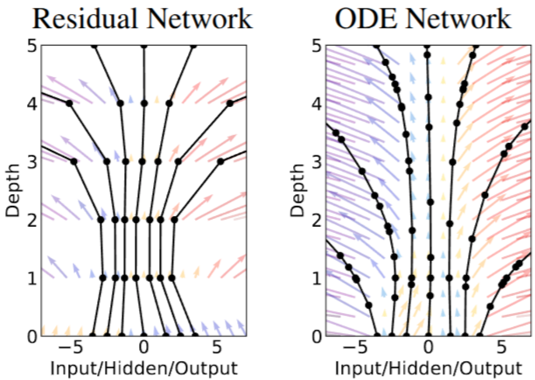


Figure 1: *Left:* A Residual network defines a discrete sequence of finite transformations. *Right:* A ODE network defines a vector field, which continuously transforms the state. *Both:* Circles represent evaluation locations.

Neural ODE for Classification

1. **Downsampling conv layers**: extract features from input image $\mathbf{x}$
2. **ODE block**: continuous feature evolution

$$\frac{d\mathbf{x}}{dt} = f_\theta(\mathbf{x}(t), t)$$

3. **Fully connected layer**: map $\mathbf{x}(t_1) \rightarrow$ class logits

ODE block is invertible.

Given $\mathbf{x}(t_1)$, integrate the ODE **backwards** in time from t_1 to t_0:

$$\mathbf{x}(t_0) = \mathbf{x}(t_1) + \int_{t_1}^{t_0} f_\theta(\mathbf{x}(t), t)\, dt$$

This gives exact **free invertibility**, an interesting property for normalizing flows!

Difficult of Bakcpropagaion in Neural ODE

Backpropagation (or reverse mode differentiation as you can call it) is difficult for continuous-depth networks.

We treat ODE as a blackbox and compute gradients using **adjoint sensitivity** method that dates back to 1962.

Adjoint senstivity method computes gradients by solving second, augmented ODE backwards in time, and is applicable to all ODE solvers.

The Adjoint Sensitivity Method: Overview

The Problem

To get trained model with parameters θ we need $\frac{d\mathcal{L}}{d\theta}$. Naively backpropping through the ODE solver requires storing all intermediate states $\Rightarrow \mathcal{O}(N_{\text{steps}})$ memory.

Adjoint State

Define the **adjoint** $\mathbf{a}(t) = \frac{\partial \mathcal{L}}{\partial \mathbf{x}(t)}$.

- $\mathbf{a}(t_1) = \frac{\partial \mathcal{L}}{\partial \mathbf{x}(t_1)}$ (from loss at output)
- $\mathbf{a}(t)$ evolves **backwards** via its own ODE:

$$\frac{d\mathbf{a}(t)}{dt} = -\mathbf{a}(t)^{\top} \frac{\partial f_\theta(\mathbf{x}(t), t, \theta)}{\partial \mathbf{x}(t)}$$

The adjoint ODE is solved **backwards in time**, recomputing $\mathbf{x}(t)$ on the fly. No need to store forward trajectory. Memory cost: $\mathcal{O}(1)$ (constant w.r.t. solver steps)!

Adjoint Method: Gradient Computation

- We can compute $\frac{\partial \mathcal{L}}{\partial \mathbf{x}(t_0)}$ by another call to ODE solver.
- This solver must run backwards starting from the initial value of $\frac{\partial \mathcal{L}}{\partial \mathbf{x}(t_1)}$.
- This may require knowing $\mathbf{x}(t)$ along its entire trajectory.
- But we can simply recompute $\mathbf{x}(t)$ backwards in time together with the adjoint starting from its final value $\mathbf{x}(t_1)$.

Computing the gradient wrt θ requires evaluating a third integral which depends on both $\mathbf{z}(t)$ and $\mathbf{a}(t)$:

$$\frac{d\mathcal{L}}{d\theta} = -\int_{t_1}^{t_0} \mathbf{a}^\top \frac{\partial f_\theta(\mathbf{x}(t), t, \theta)}{\partial \mathbf{x}(t)} dt$$

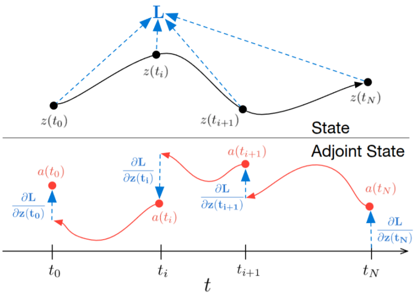


Figure 2: Reverse-mode differentiation of an ODE solution. The adjoint sensitivity method solves an augmented ODE backwards in time. The augmented system contains both the original state and the sensitivity of the loss with respect to the state. If the loss depends directly on the state at multiple observation times, the adjoint state must be updated in the direction of the partial derivative of the loss with respect to each observation.

https://arxiv.org/pdf/1806.07366

Adjoint Method: Gradient Computation

Augmented ODE (Backward in Time)

Solve the following augmented system from t_1 to t_0:

$$\frac{d}{dt}\begin{bmatrix}\mathbf{x}(t)\\ \mathbf{a}(t)\\ \frac{d\mathcal{L}}{d\theta}\end{bmatrix} = \begin{bmatrix} f_\theta(\mathbf{x}, t)\\ -\mathbf{a}^\top \frac{\partial f_\theta}{\partial \mathbf{x}}\\ -\mathbf{a}^\top \frac{\partial f_\theta}{\partial \theta}\end{bmatrix}$$

with initial conditions at t_1:

$$\mathbf{x}(t_1), \quad \mathbf{a}(t_1) = \frac{\partial \mathcal{L}}{\partial \mathbf{x}(t_1)}, \quad \left.\frac{d\mathcal{L}}{d\theta}\right|_{t_1} = \mathbf{0}$$

After integrating to t_0:

- $\mathbf{a}(t_0) = \frac{\partial \mathcal{L}}{\partial \mathbf{x}(t_0)}$ (gradient w.r.t. initial state)
- $\frac{d\mathcal{L}}{d\theta}$ accumulated across all time (gradient w.r.t. parameters)

Computation Summary

Quantity	Cost
Forward solve	1 ODE call
Backward solve	1 ODE call
Memory	$\mathcal{O}(1)$
Jacobian-vecs	via autograd

Adjoint Method: Algorithm

Algorithm 1: Neural ODE Training with Adjoint Sensitivity

Input: Parameters θ, initial state $\mathbf{x}_0$, loss $\mathcal{L}$, times $t_0 < t_1$
Output: Gradients $\frac{d\mathcal{L}}{d\theta}$, $\frac{d\mathcal{L}}{d\mathbf{x}_0}$

Forward pass:
$\mathbf{x}(t_1) \leftarrow \texttt{ODESolve}(f_\theta, \mathbf{x}_0, t_0, t_1)$;
Compute $\mathcal{L}(\mathbf{x}(t_1))$;

Backward pass (Adjoint):
Initialize $\mathbf{a}(t_1) = \frac{\partial \mathcal{L}}{\partial \mathbf{x}(t_1)}$,;
$\mathbf{g}_\theta = \mathbf{0}$;
Solve augmented ODE from t_1 to t_0: $[\mathbf{h}(t_0),\ \mathbf{a}(t_0),\ \mathbf{g}_\theta] \leftarrow \texttt{ODESolve}(\text{aug}, [\mathbf{x}_1, \mathbf{a}_1, \mathbf{0}], t_1, t_0)$;
return $\frac{d\mathcal{L}}{d\mathbf{x}_0} = \mathbf{a}(t_0)$,;
;
$\frac{d\mathcal{L}}{d\theta} = \mathbf{g}_\theta$;

The backward augmented ODE is solved with the *same* ODE solver as the forward pass, the solver is a black box, gradients flow through the **adjoint ODE**, not through solver steps.

Normalizing Flows: Background

Change of Variables

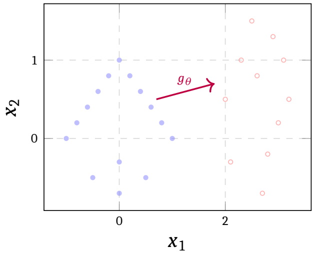
A normalizing flow maps a simple base distribution $p_z(\mathbf{z})$ (e.g., Gaussian) to a complex target $p_x(\mathbf{x})$ via an invertible map $g : \mathbf{z} \mapsto \mathbf{x}$:

$$\mathbf{x} = f(\mathbf{z}) \Rightarrow \log p_x(\mathbf{x}) = \log p_z(\mathbf{z}) - \log \left| \det \frac{\partial g}{\partial \mathbf{z}} \right|$$

Discrete NF Challenges

- Must design **special invertible architectures**
- Jacobian determinant computation: $\mathcal{O}(d^3)$ in general
- Coupling layers restrict architecture freedom

Flow: Simple → Complex



Continuous Normalizing Flows (CNF)

Instantaneous Change of Variables

For a continuous transformation $\mathbf{z}(t)$ governed by an ODE, the log-density evolves as:

$$\frac{\partial \log p(\mathbf{z}(t))}{\partial t} = -\operatorname{tr}\left(\frac{\partial f_\theta}{\partial \mathbf{z}(t)}\right)$$

This is the **instantaneous change of variables** formula (Liouville's theorem).

CNF Log-Likelihood

$$\log p(\mathbf{x}) = \log p(\mathbf{z}(t_0)) - \int_{t_0}^{t_1} \operatorname{tr}\left(\frac{\partial f_\theta}{\partial \mathbf{z}}\right) dt$$

- $p(\mathbf{z}(t_0))$: base density (Gaussian)
- Trace computable in $\mathcal{O}(d)$ via Hutchinson's estimator

★ Advantages over Discrete NF

- **Free-form** f_θ, no special invertibility constraints
- Only **trace** (not full Jacobian) needed
- Can model highly complex distributions
- Generation: integrate ODE forward; inference: backward

Hutchinson's Trace Estimator

Problem

Computing $\mathrm{tr}(J_f)$ where $J_f = \frac{\partial f_\theta}{\partial \mathbf{z}} \in \mathbb{R}^{d \times d}$ costs $\mathcal{O}(d^2)$, expensive for large d!

Hutchinson Estimator

For any random vector $\varepsilon \in \mathbb{R}^d$ with $\mathbb{E}[\varepsilon] = 0$, $\mathbb{E}[\varepsilon\varepsilon^\mathsf{T}] = I$:

$$\mathrm{tr}(J_f) = \mathbb{E}_\varepsilon \left[\varepsilon^\mathsf{T} J_f \, \varepsilon \right]$$

- $\varepsilon^\mathsf{T} J_f$ is a **vector-Jacobian product**: computable via one backward pass through f_θ, costing $\mathcal{O}(d)$
- Use Rademacher noise: $\varepsilon_i \in \{-1, +1\}$ with equal probability
- Average over K samples for low-variance estimate

Training CNFs requires only $\mathcal{O}(d)$ operations per step, making high-dimensional density estimation tractable.

Latent Neural ODEs for Time Series

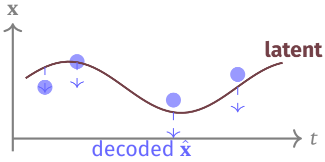
Motivation: Irregular Time Series

Standard RNNs process observations at fixed, regular intervals. Real-world data is:

- **Irregularly sampled**: medical records, sensor readings
- **Partially observed**: missing measurements
- **Multi-rate**: different sensors at different frequencies

Latent ODE Model

1. **Encoder** (ODE-RNN): infer initial latent state $\mathbf{z}(t_0)$ from observations
2. **Latent dynamics**: $\frac{d\mathbf{z}}{dt} = f_\theta(\mathbf{z}(t))$ (no explicit t-dependence)
3. **Decoder**: map latent $\mathbf{z}(t_k) \rightarrow \hat{\mathbf{x}}(t_k)$ at any time t_k



ODE-RNN Encoder

ODE-RNN

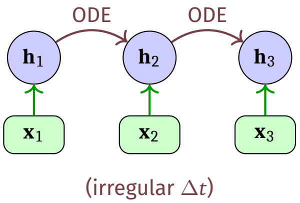
Combine an RNN (for observation updates) with an ODE (for dynamics between observations):

- At observation time t_k: **RNN update**

$$\mathbf{h}(t_k^+) = \text{RNNCell}\Big(\mathbf{h}(t_k^-),\, \mathbf{x}(t_k)\Big)$$

- Between observations: **ODE evolution**

$$\mathbf{h}(t_{k+1}^-) = \texttt{ODESolve}(f_\theta, \mathbf{h}(t_k^+), t_k, t_{k+1})$$



Handles **arbitrarily irregular** time gaps between observations naturally.

Latent ODE: Generative Model

Variational Formulation

Model observations as:

$$\mathbf{z}(t_0) \sim p(\mathbf{z}_0), \quad \mathbf{z}(t) = \mathbf{z}(t_0) + \int_{t_0}^{t} f_\theta(\mathbf{z}(s))\, ds, \quad \mathbf{x}(t_k) \sim p(\mathbf{x} \mid \mathbf{z}(t_k))$$

Train with **ELBO** (Evidence Lower Bound):

$$\mathcal{L} = \mathbb{E}_{q(\mathbf{z}_0 \mid \{\mathbf{x}(t_k)\})}\left[\sum_k \log p(\mathbf{x}(t_k) \mid \mathbf{z}(t_k))\right] - \mathrm{KL}[q(\mathbf{z}_0 \mid \cdot) \parallel p(\mathbf{z}_0)]$$

Capabilities

- Interpolate at **any time** (not just observed)
- Extrapolate into future
- Handle **missing data** gracefully
- Uncertainty quantification

Applications

- Clinical time series (ICU data)
- Physical system identification
- Financial time series
- Motion capture / trajectory modeling

Implementation with `torchdiffeq`

`torchdiffeq` Library (Chen et al.)

- PyTorch-based library for differentiable ODE solving
- Supports: `euler`, `rk4`, `dopri5` (Dormand–Prince), `adams`
- Automatic adjoint-based gradients

```python
from torchdiffeq import odeint, odeint_adjoint
import torch, torch.nn as nn

class ODEFunc(nn.Module):
    """Parametrize dh/dt = f_theta(h, t)"""
    def __init__(self, hidden_dim):
        super().__init__()
        self.net = nn.Sequential(
            nn.Linear(hidden_dim, hidden_dim),
            nn.Tanh(),
            nn.Linear(hidden_dim, hidden_dim),
        )
    def forward(self, t, h):   # Note: signature must be (t, h)
        return self.net(h)

func = ODEFunc(hidden_dim=64)
h0   = torch.randn(16, 64)          # batch_size x hidden_dim
t    = torch.tensor([0.0, 1.0])     # integration interval
# odeint_adjoint uses O(1) memory backprop
h1   = odeint_adjoint(func, h0, t, method='dopri5')[-1]
```

Complete Neural ODE Classifier

```
class NeuralODEClassifier(nn.Module):
    def __init__(self, in_dim=784, hidden=64, n_classes=10):
        super().__init__()
        self.encoder  = nn.Linear(in_dim, hidden)
        self.ode_func = ODEFunc(hidden)
        self.decoder  = nn.Linear(hidden, n_classes)
        self.t        = torch.tensor([0.0, 1.0])
    def forward(self, x):
        h0 = self.encoder(x)                        # (B, hidden)
        # Solve ODE: O(1) memory, adaptive steps
        hT = odeint_adjoint(self.ode_func, h0,
                            self.t, method='dopri5')[-1]
        return self.decoder(hT)                     # logits
# Training loop (standard)
model     = NeuralODEClassifier()
optimizer = torch.optim.Adam(model.parameters(), lr=1e-3)
criterion = nn.CrossEntropyLoss()
for x_batch, y_batch in dataloader:
    logits = model(x_batch.view(x_batch.size(0), -1))
    loss   = criterion(logits, y_batch)
    optimizer.zero_grad()
    loss.backward()      # adjoint handles backprop
    optimizer.step()
```

ODE Solver Options in `torchdiffeq`

Available Solvers

Method	Order / Type
`euler`	1st, fixed-step
`midpoint`	2nd, fixed-step
`rk4`	4th, fixed-step
`dopri5`	4/5th, adaptive
`dopri8`	8th, adaptive
`adams`	multi-step, adaptive
`bosh3`	2/3rd, adaptive

Use `dopri5` for training (accuracy), `rk4` with fixed step for fast inference.

```python
# Tight tolerances for high accuracy
h = odeint_adjoint(
    func, h0, t,
    method='dopri5',
    rtol=1e-5,
    atol=1e-6
)

# Faster inference: fixed-step
h = odeint(
    func, h0, t,
    method='rk4',
    options={'step_size': 0.1}
)
```

Advantages of Neural ODEs

Core Strengths

- **Memory Efficiency**
  Adjoint backprop: $\mathcal{O}(1)$ memory vs. $\mathcal{O}(L)$ for ResNet
- **Adaptive Computation**
  ODE solver uses more steps for complex regions; fewer steps where dynamics are simple
- **Continuous Dynamics**
  Outputs defined at *any* time $t \in [t_0, t_1]$, not just integer layers
- **Invertibility**
  Free inversion via backward ODE integration — crucial for generative models

Application-Specific Gains

- **Irregular Time Series**
  Handles variable time gaps naturally
- **Continuous Normalizing Flows**
  Exact log-likelihood with free-form architecture
- **Parameter Efficiency**
  Weights shared across all ”depths” (integration steps)
- **Physical Inductive Bias**
  Natural fit for systems governed by differential equations (mechanics, chemistry, biology)

Limitations and Challenges

Computational Challenges

- **Slow training**: ODE solving is iterative; each gradient step requires forward + backward ODE integration
- **Stiff ODEs**: dynamics with very different timescales require tiny step sizes
- **Adjoint inaccuracy**: numerical errors in backward ODE accumulate; tight tolerances needed
- **Depth-accuracy tradeoff**: coarser tolerance $\Rightarrow$ fewer steps but less accurate

Modeling Limitations

- **No crossing trajectories**: homeomorphism constraint — the ODE flow is a diffeomorphism; cannot fold or cross in latent space
- **Topology**: cannot change topology of data manifold (e.g., cannot map one cluster to two disjoint clusters without augmentation)
- **Expressiveness**: augment state space to overcome (Augmented Neural ODEs)
- **Hyperparameter sensitivity**: solver tolerance, integration range need tuning

Augmented Neural ODEs

Motivation: Overcoming Topology Limitations

Recall: an ODE flow $\phi : \mathbb{R}^d \to \mathbb{R}^d$ is a **homeomorphism** — cannot change topology. For example, points on opposite sides of a boundary cannot swap during integration (no crossing).

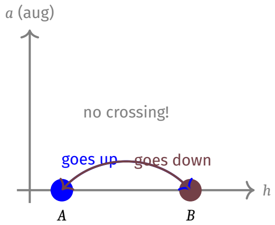
Augmented Neural ODE (ANODE)

Augment the state with p additional dimensions initialized to zero:

$$\tilde{\mathbf{x}}(t_0) = \begin{bmatrix} \mathbf{x}(t_0) \\ \mathbf{0}_p \end{bmatrix} \in \mathbb{R}^{d+p}$$

Solve the ODE in the augmented space $\mathbb{R}^{d+p}$, then read off the first d dimensions.

- Extra dimensions provide "lifting" room to route around topological barriers
- Small p (e.g., $p = 1$) often sufficient in practice



Extensions: Neural CDEs and SDEs

Neural Controlled Differential Equations (CDEs)

$$\mathbf{z}(T) = \mathbf{z}(t_0) + \int_{t_0}^{T} f_\theta(\mathbf{z}(t))\, d\mathbf{X}(t)$$

- Generalizes Neural ODEs with a **control signal** $\mathbf{X}(t)$ (the data path)
- Rigorously handles **online** time-series processing
- Handles irregular observations via the **Stieltjes integral**
- Theoretically grounded: classical CDE theory (Lyons, 1998)

Neural Stochastic DEs (SDEs)

$$d\mathbf{z}(t) = f_\theta(\mathbf{z}(t), t)\, dt + g_\theta(\mathbf{z}(t), t)\, d\mathbf{W}(t)$$

- $\mathbf{W}(t)$: Wiener process (Brownian motion)
- g_θ: diffusion / noise network
- Models **stochasticity** in dynamics
- Equivalent to infinite-width Bayesian neural network in some limits
- Score-based generative models (diffusion models) are a special case!

Connection to Score-Based Generative Models

Probability Flow ODE

Score-based diffusion models define a forward SDE that gradually adds noise to data. There exists an equivalent **deterministic ODE** (probability flow ODE) with the same marginal densities:

$$d\mathbf{x} = \left[f(\mathbf{x}, t) - \frac{1}{2}g(t)^2 \nabla_{\mathbf{x}} \log p_t(\mathbf{x})\right] dt$$

- $\nabla_{\mathbf{x}} \log p_t(\mathbf{x})$: **score function**, learned by a neural network
- This is a Neural ODE with a score-network as the dynamics!
- Enables **exact log-likelihood** computation (CNF connection)
- Allows **controllable generation** and inversion

Neural ODEs form the theoretical backbone of modern **diffusion-based generative models** (DDPM, Score SDE, Flow Matching).

TLDR

1. **Neural ODEs** replace discrete layer stacks with a continuous ODE: $\frac{d\mathbf{x}}{dt} = f_\theta(\mathbf{x}, t)$
2. **Forward pass**: call a black-box ODE solver (`ODESolve`)
3. **Backward pass**: adjoint sensitivity method — $\mathcal{O}(1)$ memory, numerically stable
4. **CNFs**: exact log-likelihood via instantaneous change of variables + trace estimator
5. **Latent Neural ODEs**: continuous-time generative model for irregular time series
6. **Augmented NODEs**: overcome topology limitations by lifting to higher dimensions
7. **Extensions**: Neural CDEs (online data), Neural SDEs (stochasticity), connect to diffusion models

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Neural ODEs unify **dynamical systems** and **deep learning**: they give continuous-depth models, constant-memory training, adaptive computation, and a principled framework for continuous-time sequence modeling and generative modeling.